IntelliCredit: FinTech Micro-Lending Risk and Alternative Credit Scoring Engine

Capstone Project, Business Analytics, Units 1, 2 and 3: Final Project Documentation

[Programme / Section] · Team [No.]

Darun R G (CB.SC.U4CSE23215)

Vimalharihar S K [Register No.]

Abhinav Juturu [Register No.]

Gowtham Reddy [Register No.]

Adithya [Register No.]

Guide: [Name, Designation]

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Repository: https://github.com/Goldmauler/intellicredit-review2

Declaration and Acknowledgement

We declare that this report and the accompanying code are our own work, carried out for the Business Analytics capstone. Sources of data, methods, software and prior studies are cited in the text and listed in the References. The use of generative-AI tools is disclosed in Section 8.3. [Signatures of all members.]

We thank [guide] for the Review 1 feedback that led us to collect and test real credit data, and the IMF, World Bank, Federal Reserve Bank of St. Louis and the US Consumer Financial Protection Bureau for open access to their data.

Abstract

Gig and platform workers in India, estimated at 7.7 million in 2020–21 and about 12 million by FY25 [16, 17], mostly have thin or no credit-bureau files, so lenders either exclude them or lend without evidence. IntelliCredit is an analytics engine that scores such borrowers from cash-flow behaviour, forecasts when portfolio defaults will rise, and explains every decision. Because individual credit records in India are confidential, borrower data (30,000 borrowers, 360,000 monthly transactions) were generated with documented, seeded logic, while all macroeconomic drivers were collected live from the FRED and World Bank APIs. In response to Review 1 feedback, five further real datasets (128,662 records from the IMF, World Bank, FRED and CFPB) were collected and used to re-test both advanced methods. A calibrated LightGBM model reaches a test AUC of 0.741. A SARIMAX model with lagged macro drivers forecasts the monthly portfolio default rate with a 12-month hold-out MAE of 0.0545 percentage points, ahead of six alternatives; on India's real bank NPL ratio a regime-aware SARIMAX attains 0.48 points against 2.04 for a naive forecast. Association rule mining yields 93 validated warning rules, and on 48,799 real loan applications 95% of mined denial rules hold in a state not used for mining. A Streamlit dashboard combines the three layers with macro-scenario approval, counterfactual paths to approval and conformal prediction intervals.

Executive Summary

Lenders to gig workers face two linked problems: they cannot tell which applicants will repay without a credit history, and they do not know when the economy will push defaults up. IntelliCredit addresses both.

- **Who will repay.** Cash-flow signals (income stability, utility bill punctuality, months on the platform, loan size relative to income) rank applicants with an AUC of 0.741, close to the 0.781 ceiling set by the data. The approval rule approves an applicant only when the predicted default probability is below the break-even level for the loan's rate and tenure.
- **When risk rises.** The monthly default rate is forecast from real Indian unemployment, GDP growth, inflation, interest-rate and exchange-rate data. Under a severe 2025 downturn expected credit loss rises from ₹1.59 crore to ₹4.32 crore, and the same rule automatically cuts approvals from 75% to 48%.
- **Why an applicant is risky.** 93 tested warning patterns, such as volatile and falling income with thin savings, flag 14% of applicants who account for 34% of defaults, including risky applicants inside the model's safest band.
- **Evidence from real data.** The forecasting and rule-mining methods were re-run on real Indian bank NPLs and on 48,799 real US loan applications and held up. The real data also showed where the simulation was wrong: COVID defaults in India were deferred by regulatory forbearance rather than spiking, and real lenders weight debt burden more heavily.

Recommended actions. Recommended actions: a two-tier decision (automatic approval only for low-risk applicants with fewer than three warning patterns), loan-size caps for new or declining-income workers, a monthly macro-adjustment of default probabilities, selective tightening on behaviour rather than on demographics in downturns, and provisioning at the baseline plus an adverse overlay of about ₹36 lakh.

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Abbreviations: AUC area under the ROC curve; ACI adaptive conformal inference; ARM association rule mining; BIC Bayesian information criterion; DPD days past due; DTI debt-to-income; ECL expected credit loss; EAD exposure at default; HMDA Home Mortgage Disclosure Act; LGD loss given default; MAE mean absolute error; MASE mean absolute scaled error; NPL non-performing loan; PD probability of default; PiT point-in-time; SARIMAX seasonal ARIMA with exogenous regressors; SHAP Shapley additive explanations; TTC through-the-cycle.

1. Introduction, Business Objectives and Literature Review

1.1 Context and scope

Digital lenders and microfinance institutions serve borrowers whom bureau scores cannot rank. About 1.4 billion adults worldwide were unbanked in 2021 [21], and India's gig workforce is growing quickly; the Economic Survey 2025-26 reports that about 40% of gig workers earn under ₹15,000 a month [17]. The stakeholders are the lender's underwriting team (approve or decline, loan size), its risk and finance teams (provisions and capital), and borrowers, who need fair and explained decisions. Scope: unsecured micro-loans of ₹5,000–₹3,00,000 over 3–24 months to individual gig workers; group lending, collections strategy and pricing optimisation are out of scope.

Table 1. Business questions and KPIs.

Analytical question	KPI (unit)	Baseline	Target / success criterion	Result
Who will default?	Test AUC	Logistic regression 0.728	≥ 0.73 and calibrated	0.741, mean PD 7.9% vs observed 7.9%
How many bad loans does the policy let in?	Default rate of approved loans (%)	7.87% (approve all)	< 6% in a neutral economy	5.09% at 60.3% approval
When will portfolio defaults rise?	12-month MAE (pp)	Naive 0.088	Beat every benchmark	0.0545 (MASE 0.079)
Which behaviour combinations signal default?	Share of rules valid on unseen borrowers	—	≥ 80% keep lift ≥ 1.3	100% of 93 rules
Do the methods hold on real data?	India NPL MAE (pp); real rules valid out-of-state	Naive 2.04	Beat naive; ≥ 80% valid	0.48; 95% of 82 rules

Figure 1 summarises the end-to-end workflow: Unit 1 (cleaning, EDA) and Unit 2 (features, models) produce the PD; Unit 3 Method 1 produces the macro multiplier and loss scenarios, Method 2 the warning rules; all three feed the dashboard, and the real-data track re-tests both Unit 3 methods.

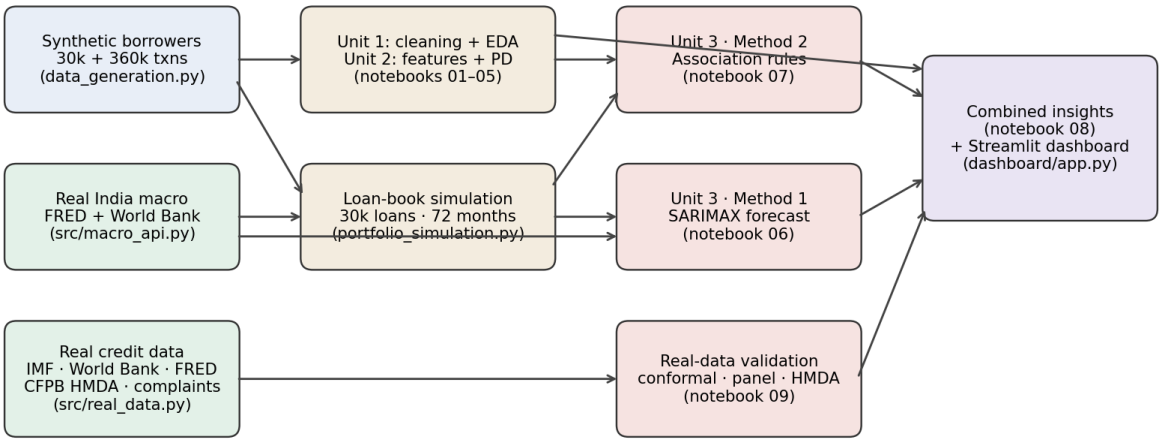


Figure 1. End-to-end workflow connecting Units 1–3, the real-data track and the dashboard. *Source: authors' diagram (scripts/build_final_report.py).*

1.2 Related analytics work and research gap

Berg et al. [1] show on about 250,000 German e-commerce purchases that a simple digital footprint predicts default as well as a bureau score (AUC 0.696 vs 0.683) and that the two combined reach 0.736. Óskarsdóttir et al. [2] add mobile-phone call records for 22,000 applicants and raise a random forest's AUC from 0.899 to 0.923. Agarwal et al. [3] report similar gains from mobile footprints at an Indian fintech. Lessmann et al. [4] benchmark 41 classifiers on credit data and find that ensembles such as random forests and gradient boosting rank above logistic regression. For portfolio-level risk, Beck, Jakubík and Piloiu [5] and Nkusu [6] link NPL ratios to GDP growth, unemployment, interest rates and exchange rates across countries. Association rules [7, 8] are widely used for basket analysis but rarely as validated, significance-tested credit warnings, where Webb [9] warns that large rule searches produce false discoveries.

The gap: (a) call-record scoring as in [2, 3] cannot be deployed in India, because RBI's Digital Lending Directions forbid lending apps from accessing contacts and call logs [15]; (b) studies rarely connect borrower-level scoring to a macro forecast that changes the approval rule month by month; (c) published rule-mining work seldom validates rules on unseen data; and (d) model-based forecast intervals assume the model is correct. IntelliCredit uses only consented cash-flow signals, links the PD to a macro forecast, validates every rule out-of-sample, and adds conformal intervals [10, 11] and actionable counterfactual explanations [12, 13].

2. Dataset Collection, Validation and Preparation

2.1 Provenance, coverage and challenges

Table 2. Data sources. No Kaggle, UCI or GitHub dataset is used. Provenance, endpoints and fetch times are logged in data/external/macro_sources.json and data/real/real_sources.json.

Dataset	Source and method	Records × variables	Period	Licence / use
Borrower profiles	Synthetic: data_generation.py (NumPy, Faker en_IN), seed 42	30,000 × 39	snapshot	own work
Monthly transactions	Synthetic, same generator	360,000 × 14	12 months	own work
India macro indicators	FRED API (CPI, call-money, 10Y G-sec, INR/USD); World Bank API (unemployment, GDP growth, lending rate)	108 months × 7 + 2 event flags	2017–2025	public, cited
Loan book and performance	Simulation of the 30,000 borrowers driven by the real macro series (src/portfolio_simulation.py, seed 2026)	30,000 loans; 250,191 loan-months	2019–2024	own work
India bank NPL ratio	IMF Financial Soundness Indicators SDMX API	63 quarters × 2	2010-Q1 → 2026-Q2	CC BY 4.0
NPL + macro panel	World Bank WDI API	5,642 country-years × 10	2000–2025	CC BY 4.0
US consumer credit	FRED (Federal Reserve delinquency and charge-off release)	158 quarters × 6	1987–2026	public domain
HMDA loan applications	CFPB HMDA Data Browser API (DE, RI, VT, 2023)	48,799 × 31	2023	US open data
Loan complaints	CFPB Consumer Complaint Database API	15,000 × 10	2020–2023	US open data

Target and dictionary. The target is loan_default_status (1 = 90+ days past due), with a 7.87% default rate. The level of observation is one borrower (Units 1–2, Method 2) or one month of the portfolio (Method 1). The data dictionary is in data/data_dictionary.txt and data/data_dictionary_review2.md; a 100-row sample is in data/sample_final_dataset.csv.

Collection challenges and fixes: World Bank series are annual, so they were interpolated to months and flagged; the lending rate stops in 2022 and the carried-forward months are excluded from history; the CFPB endpoints reject custom user-agent strings, so the library default is used; the CFPB search API pages with a search-after key, not an offset, and no longer returns narrative text, so text mining was not possible; three early IMF quarters are missing and were interpolated in logs. Every collector caches the raw response, so the pipeline also runs offline.

Synthetic generation logic: borrower attributes are drawn from documented distributions (for example log-normal income, beta-distributed utility punctuality) with correlations between income, volatility, savings and BNPL use; the default label is drawn from a logistic function of

these signals, which fixes a theoretical AUC ceiling of 0.781 (the AUC of the true default probability). To guard against circular findings, every mechanism the synthetic data relies on was re-tested on real data in notebook 09 (Section 5.3). Two held and two did not (Table 8).

2.2 Data quality, privacy and ethics

Quality checks (notebooks 01, 06, 09): no duplicate borrowers; missing values handled by KNN imputation (numeric) and mode (categorical); IQR outlier review with winsorisation of the most skewed fields; class imbalance 7.87% handled with SMOTE on training folds only; 0 missing months and 0 missing values in the 72-month modelling window. In the real data, 33 HMDA rows are identical on the 31 kept fields (kept, as they are distinct applications), and 16.5% of panel cells are missing because many countries do not report NPLs every year.

Privacy and ethics: individual credit information in India may be shared only within the credit-information system (CICRA 2005 [14]), and personal data requires purpose-specific consent under the DPDP Act 2023 [19]. The synthetic borrowers are not real people; the real datasets are aggregates (IMF, World Bank, FRED) or records already de-identified by the publisher (HMDA has no names or exact ages). No personal data is stored. A deployment would require consent through the Account Aggregator framework, data minimisation, storage in India and no access to phone contacts or call logs [15]. Known biases: the synthetic population reflects our assumptions; HMDA covers US mortgages and three states; the IMF series is system-wide, not micro-loans.

2.3 Transformations and usable dataset

General cleaning (notebook 01) is separated from model-specific preprocessing. Ordinal encoding for location tier and education, one-hot encoding for platform, log transforms of skewed amounts, and 12-month aggregation of transactions into borrower features (for example income volatility = standard deviation ÷ mean). The borrower split is stratified 80/20 (24,000 / 6,000, seed 42); scalers, WoE bins, SMOTE and rule thresholds are fitted on training rows only. Time-series models use chronological splits only. Field mapping: Unit 1 uses all raw fields; Unit 2 uses 22 application features (src/underwriting_model.py); Method 1 uses the monthly default rate and seven macro series; Method 2 uses 19 binary flags derived from Unit 2 features.

3. Exploratory Data Analysis and Visualization (Unit 1)

Notebook 02 produced 38 charts and formal tests (chi-square for categorical fields, Mann–Whitney U for continuous ones). The findings below drove the feature and rule design.

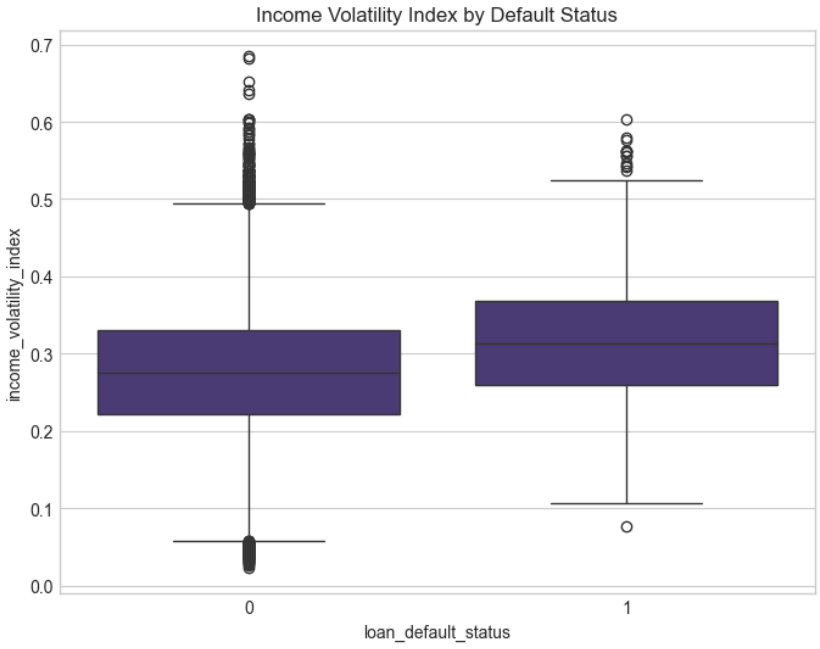


Figure 2. Income volatility index by default status: defaulters have visibly more volatile income. *Source: notebooks/02_M2_exploratory_data_analysis.ipynb, Matplotlib/Seaborn.*

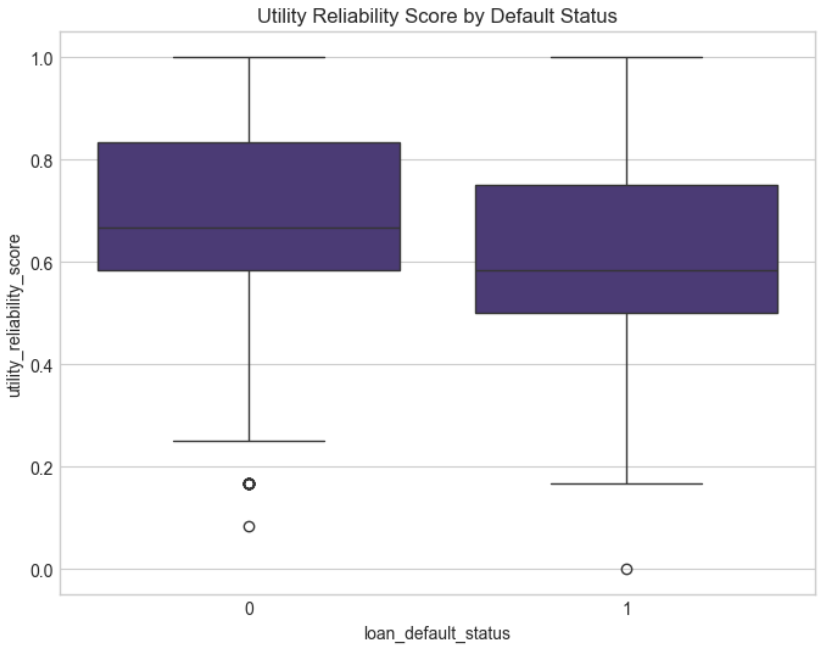


Figure 3. Utility reliability score by default status: on-time bill payment separates the classes. *Source: notebooks/02_M2_exploratory_data_analysis.ipynb.*

Income volatility, utility punctuality, savings buffer and gig tenure differ significantly between defaulters and non-defaulters; demographics (gender, location) differ little. This motivated volatility and reliability ratios as engineered features (Section 4) and the hypothesis that combinations of cash-flow stress signals, not single flags, mark the riskiest borrowers (tested in Method 2). Default rates by income quartile are flatter than by volatility, so low income alone is not treated as a risk flag in downturn policy.

4. Feature Engineering and Predictive Modelling (Unit 2)

Fourteen behavioural features were engineered (income volatility index, income trend slope, utility reliability, savings buffer ratio, debt-to-inflow ratio and others), then screened with VIF, mutual information, recursive feature elimination and Weight of Evidence / Information Value on training rows; PCA and t-SNE were used for exploration only. Five baselines and four tuned models plus voting and stacking ensembles were trained with 5-fold stratified cross-validation, with SMOTE inside each fold to avoid leakage.

Table 3. Model comparison on the same held-out test set (models/model_comparison.csv; threshold 0.5 for precision/recall).

Model (6,000 test borrowers)	AUC	Precision	Recall	F1
Logistic Regression	0.728	0.149	0.642	0.242
Logistic Regression (Tuned)	0.728	0.148	0.638	0.241
Random Forest (Tuned)	0.713	0.136	0.663	0.226
XGBoost (Tuned)	0.738	0.151	0.682	0.247
LightGBM (Tuned)	0.735	0.162	0.614	0.256
Voting Classifier (Tuned)	0.735	0.153	0.644	0.248
Stacking Classifier (Tuned)	0.739	0.000	0.000	0.000
Calibrated LightGBM (dashboard model)	0.741	—	—	—

Leakage audit (README, Review 1 fixes): preprocessing, SMOTE and WoE were refitted inside training folds; post-outcome fields were removed; there is one row per borrower, so no entity overlap. The stacking ensemble has the highest AUC but its maximum score is 0.33, so it flags nobody at 0.5; this is why the decision rule uses an economic threshold rather than 0.5. The dashboard uses LightGBM trained without class weights so that its probabilities are calibrated (mean PD 7.9% vs observed 7.9% on test).

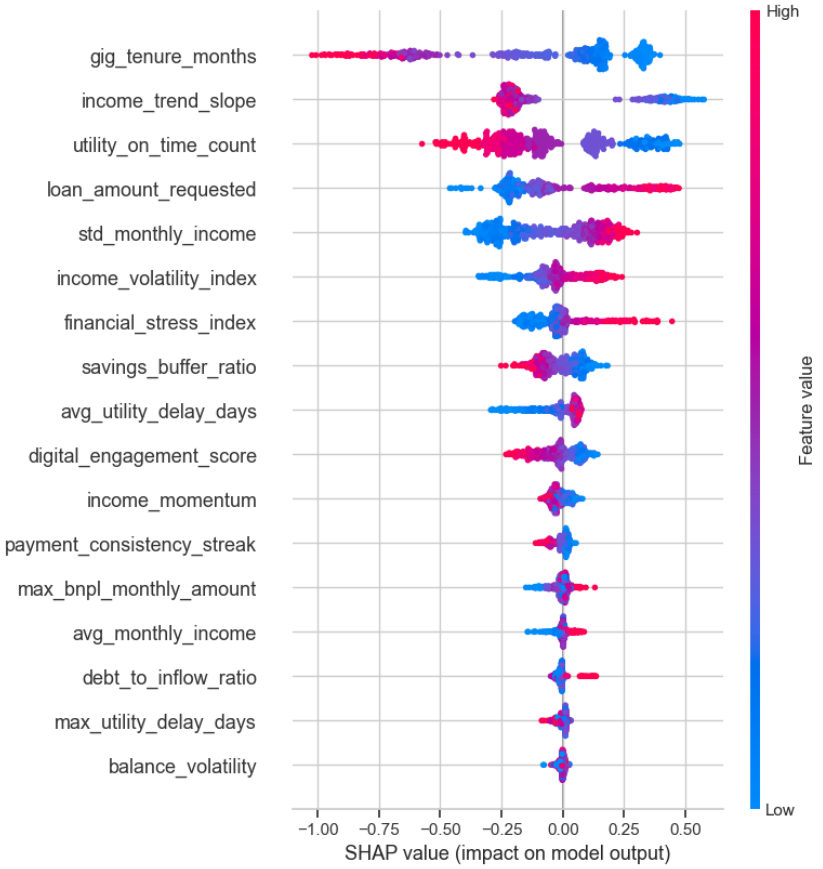


Figure 4. SHAP summary for the tuned gradient-boosting model: short gig tenure, falling income, late utility payments and large loans raise predicted risk. *Source: notebooks/05_M5_evaluation_interpretation.ipynb, SHAP TreeExplainer [18].*

Table 4. Subgroup performance of the dashboard model under the neutral break-even policy (models/subgroup_performance.csv).

Subgroup	n	Default rate	AUC	Approval rate	Default of approved
gender: Female	2429	7.7%	0.725	59.8%	5.2%
gender: Male	3442	7.9%	0.754	60.8%	5.1%
gender: Other	129	10.1%	0.646	55.0%	2.8%
location_type: Metro	1847	9.1%	0.724	57.0%	5.9%
location_type: Rural	539	6.1%	0.763	70.5%	4.2%
location_type: Semi-Urban	867	5.5%	0.766	62.3%	3.7%
location_type: Tier-1 City	1526	8.1%	0.715	58.7%	5.4%
location_type: Tier-2 City	1221	8.0%	0.765	61.4%	5.1%

Female and male applicants are approved at 59.8% and 60.8% (ratio 0.98, above the four-fifths guideline) with the same default rate among approved loans; the AUC is lower for women (0.725 vs 0.754), so discrimination is less sharp for that group, and the 'Other' group (n = 129) is too

small to judge. Rural applicants are approved more often because their default rate is lower in the data. Gender is not a model input. These checks are on synthetic data; the real-data fairness audit is in Section 5.3.

5. Advanced Analytics (Unit 3): Two Selected Methods

5.1 Method 1 — Time-series forecasting of the portfolio default rate

Business question: when will portfolio defaults rise, by how much, and what will it cost? Input: 72 monthly observations (2019-01 to 2024-12) of the default rate (new 90+ DPD loans ÷ active loans) from the macro-driven loan book, with seven real macro series. Time series is chosen because provisions and capital depend on the path of defaults over time, which a cross-sectional PD cannot give.

Preparation and implementation (notebooks/06_R2_M1_time_series_forecasting.ipynb, src/macro_api.py, src/portfolio_simulation.py): the target is modelled in logs because its variance grows with its level; drivers enter with a two-month lag because a loan reaches 90 DPD two instalments after its first miss; train 2019–2023 (60 months), test 2024 (12 months). STL gives trend strength 0.46 and seasonal strength 0.10. ADF ($p = 0.082$) and KPSS ($p = 0.01$) indicate non-stationary levels. Granger tests [20] (minimum p over lags 1–4) are significant for unemployment (0.0003), GDP growth (0.004), the call-money rate (0.010) and rupee depreciation (0.046), borderline for CPI (0.051). Seven forecasters were compared; the SARIMAX driver set was chosen by BIC on training data: SARIMAX$(1,0,1)\times(1,0,0)_{12}$ with the five lagged drivers and two COVID dummies (BIC 37.0).

Table 5. Method 1 evaluation: 12-month hold-out (2024) and rolling-origin backtest (models/ts_model_comparison.csv, ts_backtest_summary.csv).

Model	MAE (pp)	RMSE (pp)	MAPE %	MASE	Backtest MAE (40 origins)
SARIMAX + macro	0.0545	0.0677	16.4	0.079	0.124
Gradient boosting	0.0650	0.0728	21.9	0.094	—
Holt-Winters (ETS)	0.0747	0.1021	23.9	0.108	0.208
Naive	0.0881	0.1124	24.2	0.127	—
SARIMA	0.0881	0.1023	29.5	0.127	0.176
12-month moving average	0.1158	0.1317	42.3	0.167	—
Seasonal naive	0.1236	0.1641	40.4	0.178	0.351

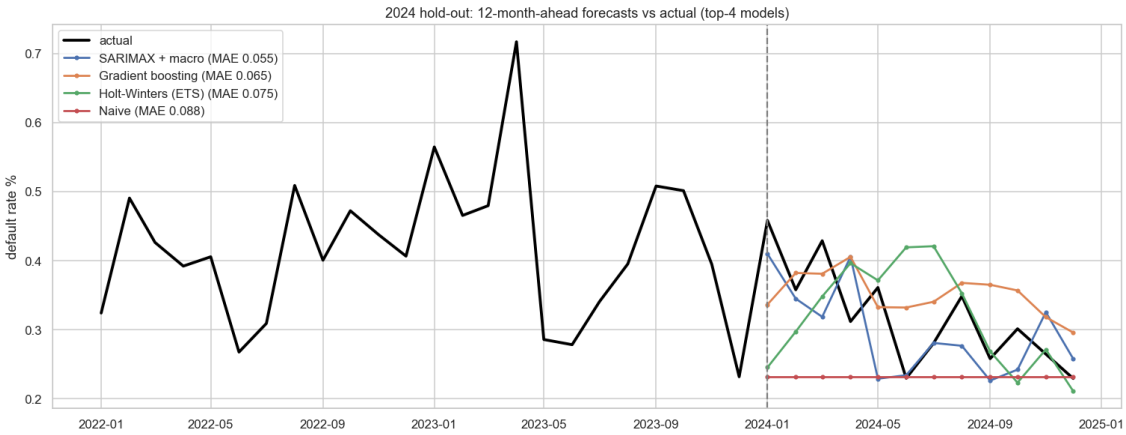


Figure 5. 2024 hold-out: 12-month-ahead forecasts against the actual default rate. *Source: notebooks/06_R2_M1_time_series_forecasting.ipynb, statsmodels.*

Interpretation: a 1-point rise in unemployment two months earlier raises the default rate by 20.8%, CPI inflation by 15.3%, the call-money rate by 10.4%; GDP growth lowers it by 1.9% per point; lockdown months multiply it by 5.65. Residuals pass Ljung–Box ($p = 0.12$ at lag 6, 0.28 at lag 12) and Jarque–Bera ($p = 0.31$). The 80% interval covered 11 of 12 test months. The 30+ DPD rate nowcasts defaults two months ahead (R^2 0.76, 2024 MAE 0.038 pp). For 2025, expected credit loss is ₹158.9 lakh (baseline), ₹194.6 lakh (adverse) and ₹432.5 lakh (severe). Limitations: the forecast is conditional on the macro path; in the 2021 second-wave months of the backtest SARIMAX (MAE 0.381) was worse than ETS (0.276); bias is −0.023 pp, a slight under-forecast; annual World Bank series are interpolated.

5.2 Method 2 — Association rule mining of delinquency warning patterns

Business question: which combinations of behaviours occur together in borrowers who default far more often than chance? Input: the 24,000 training borrowers, each turned into a basket of 19 binary flags (average 5.22 per borrower) whose quantile thresholds were computed on training rows only; validation on the 6,000 test borrowers. Rules were chosen because underwriters need readable, multi-factor red flags that a single probability does not give.

Implementation (notebooks/07_R2_M2_association_rule_mining.ipynb, src/risk_rules.py): Apriori [7] and FP-Growth [8] at minimum support 0.5% found the same 2,852 frequent itemsets (Apriori was faster on these 20 dense items, about 0.6 s vs 1.0–1.1 s). From 21,684 rules, 242 have consequent {DEFAULT}; 170 pass lift ≥ 1.5, support ≥ 0.5% and a Bonferroni-corrected

one-sided Fisher exact test; 93 remain after a productive-rule filter that requires each rule to beat all of its sub-rules by 5%.

Table 6. Top validated warning rules on unseen borrowers (base default rate 7.87%).

Rule ⇒ DEFAULT	Train lift	Test coverage	Test default rate	Test lift
Declining income + volatile income + low savings	2.57	3.85%	24.7%	3.14
Volatile income + low savings + new gig worker	2.47	3.75%	24.4%	3.11
Declining income + large loan vs income + low savings	2.30	2.75%	24.2%	3.08
Declining income + low savings + new gig worker	2.79	4.10%	23.6%	3.00

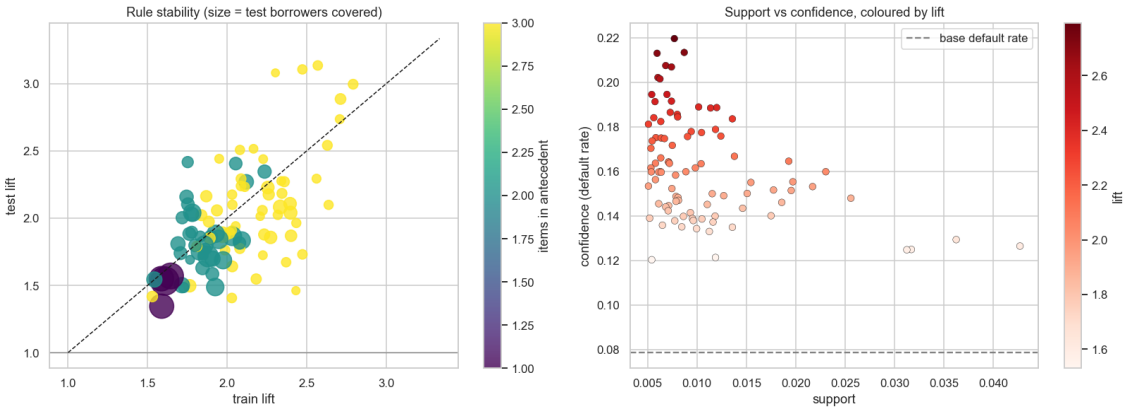


Figure 6. Train vs test lift of the 93 rules: all keep lift ≥ 1.3 on unseen borrowers (correlation 0.61, median lift 2.03 → 1.89). *Source: notebooks/07_R2_M2_association_rule_mining.ipynb, mlxtend.*

Interpretation: single flags are weak (best lift 1.65), but three-flag combinations of income instability and thin savings reach a test default rate near 25%. Applicants triggering three or more rules are 14% of test applicants and 34% of defaults. Under macro stress, income volatility adds 4.45 percentage points of extra default risk while low income adds none. Limitations: the default rate by rule count rises overall but not monotonically (17.4%, 13.8%, 11.3% for 3, 4 and 5 rules); the rule-count AUC is 0.645 when rules are ranked on training lift (0.657 when ranked on test lift).

5.3 Re-testing both methods on real data (notebook 09)

Table 7. Method 1 on India's real bank NPL ratio (IMF FSI), trained on 2010-Q1 to 2022-Q4.

Model (India NPL ratio, 14 unseen quarters)	MAE (pp)	MAPE %	MASE
SARIMAX (BIC choice)	0.482	18.8	1.262
Holt damped trend (ETS)	0.913	37.9	2.391
ARIMA(1, 1, 1)	1.030	43.1	2.698
Naive (last value)	2.036	85.7	5.332
Drift	2.606	110.7	6.824

BIC selected a SARIMAX with three regulatory regime ramps (RBI Asset Quality Review recognition +8.0% per quarter, IBC clean-up −5.4% per quarter, COVID standstill). Macro drivers did not improve BIC once regimes were included: in India, recognition rules moved reported NPLs more than the business cycle did.

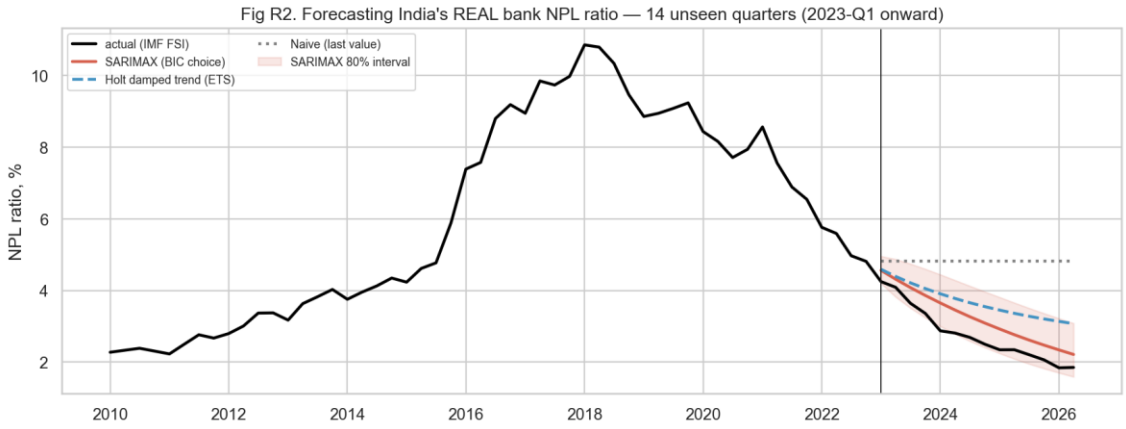


Figure 7. Forecasting India's real bank NPL ratio for 14 unseen quarters. *Source: notebooks/09_R3_real_data_validation.ipynb; IMF FSI.*

Conformal intervals: on 94 rolling one-step forecasts of real US consumer-loan delinquency, the adaptive conformal 80% band [11] covered 81% and the 90% band 90%, against 72% and 83% for the model's Gaussian bands.

Macro mechanism: a two-way fixed-effects panel of 140 countries (2,021 country-years, standard errors clustered by country) gives ΔNPL coefficients of +0.280 for the change in unemployment ($p = 0.015$), −0.111 for GDP growth ($p = 0.048$) and +0.080 for inflation ($p = 0.002$), the same signs as the simulation.

Method 2 on real loans: on 39,451 HMDA applications from Delaware and Rhode Island, Apriori produced 82 productive DENIED rules; 95% keep lift ≥ 1.3 on 9,348 Vermont applications never used for mining, and the rule-count score has AUC 0.707 there. Debt-to-income is the most cited real denial reason (43% of denials). After the risk flags, Black and Asian applicants still have about 1.7 times the odds of denial ($p < 0.001$); public HMDA has no credit scores, so this is a signal for review rather than proof of discrimination [22].

Table 8. Synthetic-vs-real fidelity scorecard.

Claim	Synthetic	Real	Verdict

Macro direction: unemployment ↑ ⇒ defaults ↑, GDP ↑ ⇒ defaults ↓	yes (nb 06)	panel β: Δunemp +0.280, GDP -0.111 (140 countries)	CONFIRMED
Default level falls 2019 → 2024 as the economy recovers	0.82% → 0.32%	9.0% → 2.7% (corr 0.64)	CONFIRMED
COVID 2020 default spike	×2.3 (2020 vs 2019)	×0.89 — moratorium & standstill hid it	CONTRADICTED → add forbearance regime
Debt burden as a risk driver	loan-to-income flag lift 1.12	DTI ≥ 50% flag lift 2.55 (HMDA denials)	UNDER-WEIGHTED → recalibrate generator
Multi-flag rules beat single flags & generalise out-of-sample	93 rules, 100% hold on test	82 rules, 95% hold in an unseen state	CONFIRMED

6. Comparative Analysis and Business Impact

6.1 Comparison of the two Unit 3 methods

Table 9. Method comparison. Their metrics are not comparable numerically and are not ranked against each other.

	Method 1: time series	**Method 2: association rules**
Question	When and how much will portfolio defaults move?	Which behaviour combinations mark risky borrowers?
Input	72 monthly rates + 7 real macro series	24,000 borrowers × 19 binary flags
Output	12-month forecast, scenarios in ₹, PD multiplier	93 readable warning rules, rule-count score
Validation	Chronological hold-out, 40-origin backtest, residual tests, interval coverage	Unseen-borrower lift, Bonferroni tests, productive-rule filter
Strength	Links defaults to the economy; supports provisioning	Explains decisions; finds risk inside low-PD bands
Limitation	Conditional on macro path; few observations	Rules give reasons, not ranking (AUC 0.645–0.657)

Synthesis: the methods are complementary. Within PD band A, applicants with six or more rules default at 5.6% against 1.8% with none, so rules add information the PD lacks. The time-series multiplier makes the same break-even rule approve 75.1% of applicants in the baseline 2025 economy and 47.8% in a severe one, while the approved default rate falls from 6.10% to 3.98%. The macro-conditioned rules join the two: in downturns the dangerous flags are behavioural.

6.2 Comparison with existing studies

Our test AUC (0.738 tuned XGBoost; 0.741 calibrated LightGBM) is in the range reported by Berg et al. [1] (0.683–0.736) and below Óskarsdóttir et al. [2] (up to 0.923). Direct comparison is not valid: their labels are real repayments, ours are generated with a known ceiling of 0.781; their features include bureau scores or call records that Indian rules forbid for app lenders [15]. Our

ranking of ensembles above logistic regression agrees with Lessmann et al. [4]. On the macro side, the signs we find on real data match the cross-country evidence of [5, 6]. Our distinct contributions are the link from a macro forecast to an approval rule, out-of-sample validated rules, and conformal intervals. Remaining gaps: no real micro-loan repayment labels and no live deployment. Full benchmark tables are in research/BENCHMARK_TABLE.md.

6.3 KPI-based business impact and scenarios

Table 10. Scenario impact. Approvals are measured on the 6,000 test applicants; credit losses are model forecasts.

Scenario (2025)	PD multiplier	Approval	Default of approved	Expected credit loss	vs baseline
Neutral economy	1.00	60.3%	5.09%	—	—
Baseline	0.57	75.1%	6.10%	₹158.9 lakh	—
Adverse (unemployment +1.5 pp, GDP −3 pp)	0.69	70.7%	5.73%	₹194.6 lakh	+₹35.6 lakh
Severe (COVID-style)	1.53	47.8%	3.98%	₹432.5 lakh	+₹273.6 lakh

Assumptions: LGD 65%, cost of funds 9%, average exposure at default ₹111,589, book growth 1.38% per month. Measured outcomes (approval and default rates on the test set) are separated from forecasts (2025 losses). Sensitivity: the spread between baseline and severe losses (₹273.6 lakh) is the capital the lender should be able to absorb; the result depends on the macro path, the LGD assumption and the simulation's COVID response, which real Indian data suggest overstates short-run defaults when regulators grant forbearance. Priorities by impact and feasibility: (1) two-tier decision with rules, (2) monthly macro multiplier, (3) loan-size caps, (4) selective downturn tightening, (5) provisioning overlay.

7. Interactive Application, Figure Index and Reproducibility

7.1 Dashboard

Platform: Streamlit (dashboard/app.py), run locally with streamlit run dashboard/app.py; a standalone Plotly report is in reports/interactive_dashboard.html. Six tabs: Underwriter decision engine (live form, PD, point-in-time PD, break-even, TreeSHAP reasons, triggered rules, counterfactual path to approval), Dataset explorer (filters by band, location, platform, outcome and rules hit, with dynamic KPIs and charts), Portfolio outlook (scenario forecasts and ₹ losses), Warning rules (lift and coverage sliders), Real-data reality check (India NPL forecasts,

conformal band, HMDA rule explorer, fidelity scorecard) and Model card. Computed live: features, PD, SHAP, decision, rules and recourse; precomputed: model, rules, scenario forecasts and notebook-09 results.

Tests: Streamlit AppTest renders every tab without exceptions, including the severe scenario with the high-risk preset, which exercises the recourse search. Known limitations: the recourse search uses a fixed grid (1,134 combinations); the dashboard is not deployed publicly. [Insert screenshots of the main state and a filtered state from reports/screenshots/.]

7.2 Figure index and reproduction map

Table 11. Figure index.

Figure	Caption	Reproduce with
1	End-to-end workflow connecting Units 1–3, the real-data track and the dashboard.	authors' diagram (scripts/build_final_report.py)
2	Income volatility index by default status: defaulters have visibly more volatile income.	notebooks/02_M2_exploratory_data_analysis.ipynb, Matplotlib/Seaborn
3	Utility reliability score by default status: on-time bill payment separates the classes.	notebooks/02_M2_exploratory_data_analysis.ipynb
4	SHAP summary for the tuned gradient-boosting model: short gig tenure, falling income, late utility payments and large loans raise predicted risk.	notebooks/05_M5_evaluation_interpretation.ipynb, SHAP TreeExplainer [18]
5	2024 hold-out: 12-month-ahead forecasts against the actual default rate.	notebooks/06_R2_M1_time_series_forecasting.ipynb, statsmodels
6	Train vs test lift of the 93 rules: all keep lift ≥ 1.3 on unseen borrowers (correlation 0.61, median lift 2.03 → 1.89).	notebooks/07_R2_M2_association_rule_mining.ipynb, mlxtend
7	Forecasting India's real bank NPL ratio for 14 unseen quarters.	notebooks/09_R3_real_data_validation.ipynb; IMF FSI

7.3 Code and environment

Repository: https://github.com/Goldmauler/intellicredit-review2 (README with project tree, requirements.txt, Python 3.12). Reproduce: python data_generation.py; python -m src.macro_api; python -m src.portfolio_simulation; python -m src.real_data; execute notebooks 01–09 in order; python -m src.underwriting_model; streamlit run dashboard/app.py; python scripts/build_final_report.py. Folders: data/ (raw, processed, external, portfolio, real), notebooks/, src/, models/, dashboard/, reports/, research/. No secrets are stored; all APIs used are keyless. Seeds: 42 (data, models), 2026 (loan book).

8. End-to-End Validation, Traceability and Contributions

8.1 Completion and validity

Table 12. Objective-to-evidence traceability matrix.

Objective	Evidence	Status
Score thin-file borrowers	notebooks 03–05; Table 3; AUC 0.741, calibrated	met
Forecast portfolio defaults	notebook 06; Table 5; MAE 0.0545 pp	met
Find validated warning patterns	notebook 07; Table 6; 93 rules, 100% valid	met
Combine into decisions	notebook 08; Table 10; dashboard	met
Respond to the 'use real data' feedback	notebook 09; Tables 7–8; 5 real datasets	met
Real micro-loan repayment labels	not legally obtainable	not met (future work)

Threats to validity: borrower labels are synthetic, so absolute performance is bounded by our generator; the macro mechanism was simulated and is only externally supported by the real panel; the real datasets are not micro-loans; the short monthly series limits model complexity. Reproducibility: all code is seeded and notebooks re-execute with identical model outputs (checked on 6 Nov 2026).

8.2 Project tracking and individual work

Board: GitHub issues on the team repository (19 tasks: 10 Review 1, 9 Review 2) with stages Backlog, To Do, In Progress, Review/Testing and Completed; task history is in reports/project_management_and_contributions.md. [Insert board screenshot.]

Table 13. Individual contributions (as recorded in the README; confirm before submission). Individual marks may vary with participation and explanation.

Member	Main contribution	Evidence
Darun R G	Data generator, Review 1 pipeline, dataset validation, project board	data_generation.py, notebooks 01–05, DATASET_VALIDATION_SUMMARY.md
Vimalharihar S K (Goldmauler)	Method 1 and Method 2, macro APIs, real-data track, recourse	notebooks 06, 07, 09; src/macro_api.py, real_data.py, recourse.py
Abhinav Juturu	EDA, loan-book simulation, Method 1 data preparation	notebook 02, src/portfolio_simulation.py
Gowtham Reddy	Cleaning, Streamlit dashboard	notebook 01, dashboard/app.py
Adithya	Evaluation, combined insights, report and slides	notebooks 05, 08, reports/

8.3 AI use and originality declarations

[To be completed and signed by the team.] Generative-AI tools were used: Claude (Anthropic), for drafting code for the real-data collectors, notebook 09, the recourse module, the dashboard tab and this report draft, for checking documents against notebook outputs, and for collecting and verifying literature. All generated code was executed and its outputs were checked against the saved results; numbers in this report were read from those outputs. [List any other tools and the parts each member wrote or verified.] Turnitin similarity: [__ %] (limit 15%).

9. Conclusion, Recommendations and Roadmap

All five objectives in Table 1 that could be met legally were met (Table 12). The engine ranks thin-file borrowers close to the data's ceiling, forecasts portfolio defaults better than six alternatives, produces rules that hold on unseen borrowers, and both advanced methods carried over to real data.

Table 14. Prioritised recommendations.

Recommendation	Expected KPI effect	Owner	Feasibility
Two-tier decision: auto-approve bands A–B with < 3 rules	Catches 34% of defaults in 14% of applicants for review	Credit policy	High
Monthly macro multiplier from the forecast	Approval adjusts 75% → 48% between baseline and severe	Risk analytics	High
Loan-to-income cap for tenure < 6 months or falling income	Fewer early defaults among new workers	Product	High
Tighten on behaviour, not demographics, in downturns	Protects inclusion of low-income applicants	Credit policy	Medium
Provision baseline ECL + adverse overlay (~₹36 lakh)	Capital ready for a moderate downturn	Finance	High
Track regulatory regimes in any India NPL forecast	India NPL MAE 0.48 vs 0.91 pp without regimes	Risk analytics	High

Roadmap. Immediate: merge the real-data track and present it. Medium term: add a forbearance regime to the loan-book simulation and raise the weight of debt burden in the generator; report conformal intervals on the portfolio forecast. Longer term: a consented gig-worker survey pilot to compare income volatility and BNPL use with the generator, and testing the rules and model on a partner lender's repayment data. Success checks: rule lift stays ≥ 1.3 on new data and PD calibration error stays below one percentage point.

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Appendices

Appendix A: data dictionaries (data/data_dictionary.txt, data/data_dictionary_review2.md), generation settings (data_generation.py, seed 42) and provenance logs (data/external/macro_sources.json, data/real/real_sources.json).

Appendix B: additional figures and tuning tables (notebooks 02–07; reports/figures/).

Appendix C: installation and dashboard guide (README, How to Run).

Appendix D: board evidence, contribution records, AI declaration and similarity report [to be attached].